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## **A Multi-Scale Health Impact Assessment of Air Pollution Over the 21st Century**

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## **Abstract:**

**Background:** Ozone and PM<sub>2.5</sub> are current risk factors for premature death all over the globe. In coming decades, substantial improvements in public health may be achieved by reducing air pollution. To better understand the potential of emissions policies, studies are needed that assess possible future health impacts under alternative assumptions about future emissions and climate across multiple spatial scales. **Method:** We used consistent climate-air-quality-health modeling framework across three geographical scales (World, Europe and Ile-de-France) to assess future (2030-2050) health impacts of ozone and PM<sub>2.5</sub> under two emissions scenarios (current legislation emissions, CLE, and maximum feasible reductions, MFR). **Results:** Consistently across the scales, we found more reductions in deaths under MFR scenario compared to CLE. 1.5 [95% CI: 0.4, 2.4] millions of CV deaths could be delayed each year in 2030 compared to 2010 under MFR scenario, 84% of which would occur in Asia, especially in China. In Europe, the benefits under MFR scenario (219 000 CV deaths) are noticeably larger than those under CLE (109 000 CV deaths). In Ile-de-France, under MFR more than 2 830 annual CV deaths associated with PM<sub>2.5</sub> changes could be delayed in 2050 compared to 2010. In Paris, ozone-related respiratory mortality should increase under both scenarios. **Conclusion:** Multi-scale HIAs can illustrate the difference in direct consequences of costly mitigation policies and provide results that may help decision-makers choose between different policy alternatives at different scales.

**Keywords:** health impact assessment, air quality modeling, emissions scenarios, cardiovascular disease, particulate matter

## **Highlights:**

- We assessed impact of air pollution on health in 2030-2050.
- We used two ECLIPSE emissions scenarios (CLE, MFR) on three geographical scales.
- We projected larger impacts on CV and respiratory mortality under the MFR scenario.
- The impacts can be larger on finer scale, due to a resolution or a model choice.
- Multi-scale HIA can provide relevant results for decision-makers.

## 1. Introduction

Numerous epidemiological studies have shown that exposure to outdoor air pollution (OAP) leads to adverse health outcomes, including increases in cardiovascular and respiratory mortality and morbidity (Beelen et al. 2013, Hoek et al. 2013, Krewski et al. 2009, Pope III et al. 2002, Pope III & Dockery 2006, Raaschou-Nielsen et al. 2013). The International Agency for Research on Cancer (IARC) recently classified OAP (including particulate matter (PM) as a major component of it) as carcinogenic to humans (Group 1) (IARC 2013). While individual-level health risks associated with OAP may be low compared to other risk factors, the overall population-wide public-health impact can be very large. For instance, according to the World Health Organization's Global Burden of Diseases (Lozano et al. 2012), in 2010, annually, 3.22 million people died prematurely because of outdoor PM, and 0.15 million because of ozone (O<sub>3</sub>).

Health impact assessments (HIAs) have been extensively used to quantify the public-health impact of OAP and to help decision-makers understand the benefits that would be associated with an improved air quality (Medina et al. 2013). With the development of air quality and climate models, there is a growing interest in quantifying the future health-impacts of air pollution, taking into account trends in emissions, air-pollution policies and climate change. Joint policies to reduce emissions of air pollutants and greenhouse gases are promising, as climate and air pollutants are associated through dynamic processes at multiple scales, from common emissions sources to their chemical and physical interactions in the atmosphere (Colette et al. 2012b, Jacob et al. 2009). HIA can help to understand the health benefits that could potentially be achieved under different scenarios of air pollution emissions and of climate change, and therefore encourage synergies and limit trade-offs between mitigation of climate change and mitigation of air pollution.

Several HIAs examining future air pollution-related health impacts have been published using a range of climate and air quality models, scenarios, and spatial/temporal scales (Anenberg et al. 2012, Heal et al. 2012, Post et al. 2012, Orru et al. 2013). The choice of appropriate spatial and temporal scale to make such exercise meaningful to decision-makers is a crucial issue. In this paper, we present and compare HIAs of future air pollution carried out in a consistent way across three different geographical scales: the World, Europe and the French Ile-de-France (IdF) region. Our focus here is on health impacts rather than on modeling issues. The objective is two-fold: for scientists, it allows comparing the results across scales; for decision-makers, it allows putting the results into a large perspective illustrating that air quality and climate change are not only global, but also local issues.

## 2. Materials and Methods

### 2.1. General framework and scenarios

To quantify the health impacts of OAP on three geographical scales we used alternative scenarios for air pollutant emissions, and climate change (Figure 1). These emissions scenarios were inputs to climate and chemistry models, which were coupled across the scales.

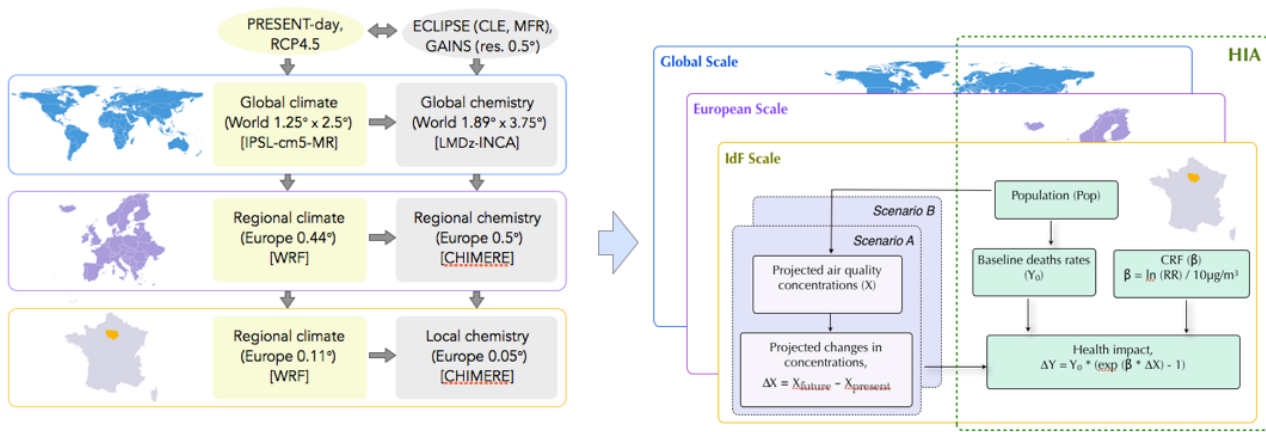


Figure 1. The scheme of air pollution and climate modeling (left) and HIA method (right) framework: at each scale a climate model feeds a chemistry model to assess future concentrations of air pollutants, which are used in the HIA analysis. In the round brackets: (Geographical area of the modeling and the size of the grid cells). In the square brackets: [name of the model]

We used two air pollution emissions scenarios consistent across the scales, the “Current Legislation” (CLE) and the “Maximum Feasible Reduction” (MFR). These scenarios were developed in the framework of the ECLIPSE (Evaluating the Climate and Air Quality Impacts of Short-Lived Pollutants) project (Amann et al. 2013, Klimont et al. 2013a, Klimont et al. (In preparation for ACPD)). They include both climate and regional air quality policies for the emissions of air pollutants. The CLE scenario assumes the existing air quality legislation is fully implemented and enforced. The MFR assumes that all technologically feasible emissions reduction measures are implemented. It is computed using the lowest range of emissions factors in the GAINS (Greenhouse Gas and Air Pollution Interactions and Synergies) model (Amann et al. 2011) but ignores any non-technical measures. It has to be noted that while indeed the air quality policies should be mostly enforced until 2030 (Amann et al. 2013) there still might be significant changes, that is, in some world regions the activity data (e.g. economic growth) might lead to strong increases in air pollution and rebound air pollution effects unless stronger legislation is introduced. The CO<sub>2</sub> trajectory of ECLIPSE is similar (at a global level) to the RCP6.0 but the emission of pollutants develop differently, typically the RCPs reduce emissions much quicker than CLE in ECLIPSE/GAINS and the reasons of that were discussed in the Amann et al. (2013).

From the ECLIPSE database we used anthropogenic emissions. We used Lathière et al. (2006) for biogenic emissions, Koffi et al. (2010) for the aviation and shipping emissions, and van der Werf et al. (2010) for forest and savannah burning emissions. Other source emissions were calculated within the air pollution models. For each scale, we performed two sets of continuous simulations, one for the present time and another one for the two future ECLIPSE-V4a projections. The air pollutant emissions used for the three time frames are those of 2010, 2030 and 2050 and for the sake of simplification we will use these labels in the following of the paper.

## 2.2. Climate modeling

At the global scale, we used meteorological data from the European Center for Medium-Range Weather Forecasts (ECMWF) reanalysis. To isolate the impact of anthropogenic emissions, we performed simulations under present-day climate conditions (2005-2006). The impact of future climate change on particles and chemistry was not taken into account; we used the results of 2006 meteorological conditions in all simulations. The role played by climate change and the impact of nitrate on the indirect aerosol radiative forcing of climate will be investigated in a forthcoming study.

At the European scale, the future evolution of climate was assessed using the IPSL-cm5-MR member of CMIP5 global-scale model (Taylor et al. 2012) downscaled spatially at a resolution of  $0.44^\circ$  with the WRF meteorological model (Skamarock et al. 2008) used as regional climate model. The simulations used are the low-resolution IPSL-INERIS member of the Euro-Cordex ensemble (Jacob et al. 2013). The RCP4.5 radiative forcing pathway was used to drive the climate simulations.

The meteorology for IdF derived from the high-resolution IPSL-INERIS member of the Euro-Cordex ensemble at  $0.11^\circ$  ( $\sim 10\text{km}$ ) resolution over Europe.

### 2.3. Air pollution Modeling

#### 2.3.1. Global Scale

The future evolution of global atmospheric composition has been simulated with the LMDz-INCA global chemistry-aerosol-climate model coupling on-line the LMDz (Laboratoire de Météorologie Dynamique, version 5) General Circulation Model (GCM) (Hourdin et al. 2006) and the INCA (INteraction with Chemistry and Aerosols, version 4) model (Hauglustaine et al. 2004). In the present configuration, the model includes 39 hybrid vertical levels. The horizontal resolution is  $1.89^\circ$  in latitude and  $3.75^\circ$  in longitude, which is equivalent to 201 km in the mid-latitude. For a more detailed description and an extended evaluation of the GCM we refer to Hourdin et al. (2006). INCA includes a state-of-the-art  $\text{CH}_4\text{-NO}_x\text{-CO-NMHC-O}_3$  tropospheric photochemistry (Hauglustaine et al. 2004, Folberth et al. 2006). The tropospheric photochemistry and aerosol scheme used in this model version is described through a total of 123 tracers including 22 tracers to represent aerosols. For aerosols, the INCA model simulates the distribution of aerosols with anthropogenic sources such as sulfates, nitrates, black carbon, particulate organic matter, as well as natural aerosols such as sea-salt and dust (Schulz 2007, Balkanski 2011, Hauglustaine et al. 2014).

#### 2.3.2. European Scale

To model the impact of future emissions on air pollution concentrations in Europe at a regional scale, we used the CHIMERE chemical transport model (Menut et al. 2013a) that computes the fate of primary and secondary pollutants (including secondary inorganic and organic aerosols). The model has been widely used for research applications involving historical air quality (Bessagnet et al. 2004, Bessagnet et al. 2008, Vautard et al. 2005) as well as operational air quality forecasting (Rouil et al. 2009, Zhang et al. 2012). It has been benchmarked in terms of  $\text{O}_3$  and PM to other existing tools in various model inter-comparison exercises (van Loon et al. 2007, Vautard et al. 2009), some of which focused on long term simulations (Colette et al. 2011, Colette et al. 2012a). We used the model outputs at  $0.5^\circ$  ( $\sim 50\text{km}$ ) resolution covering Europe.

The chemical boundary conditions were obtained from a 10-year average of chemical composition over Europe simulated with the global LMDz-INCA model for the present day (Szopa et al. 2012).

#### 2.3.3. Ile-de-France Scale

The Ile-de-France region (IdF) is located at  $1.25^\circ\text{--}3.58^\circ$  east and  $47.89^\circ\text{--}49.45^\circ$  north with a population of approximately 11.7 million, more than two million of which live in the city of Paris. The area is situated away from the coast and is characterized by uniform and low topography, not exceeding 200 m above the sea level. To model air quality in the IdF region we used the CHIMERE model at  $0.05^\circ$  ( $\sim 4\text{km}$ ) horizontal resolution in a  $156\times 128$  km domain covering IdF.

Air pollution emissions for IdF were available at hourly intervals with a spatial resolution of  $1\times 1$  km (aggregated to  $4\text{km}$ ). The emissions dataset was compiled by local experts using a variety of city-specific information integrating a number of anthropogenic activities in the region (AIRPARIF 2012). The spatial

allocation of emissions was completed using proxies such as high-resolution population maps, road network and the location of industrial units. Boundary and initial conditions for the IdF simulations were taken from the European scale simulation.

In order to maintain consistency and to be able to compare the results amongst the scales we harmonized the local inventory in terms of annual emissions fluxes with the ECLIPSE estimates used in the regional model. To achieve our goal we synchronize the emission totals of the ECLIPSE and AIRPARIF inventories. We develop a new version of the AIRPARIF inventory by scaling its total annual per pollutant emissions to match the ECLIPSE annual quantified fluxes. The spatial representation and temporal patterns (monthly, weekly, diurnal) of the final inventory are therefore adopted by AIRPARIF. To provide future estimates of emissions in IdF under the CLE and MFR scenarios we employed an equivalent procedure.

## 2.4. Population and Health Data

We used gridded world population data, country boundaries and coastlines from the Center for International Earth Science Information Network (CIESIN)/Columbia University, and Centro Internacional de Agricultura Tropical (CIAT) (2005). To avoid discrepancy between the gridded population estimates and the population counts from the WHO mortality database (2011), we adjusted the gridded population data to the population older than 15 for the year 2008. The population counts by commune in IdF were obtained from the Institut national de la statistique et des études économiques (INSEE, 2009).

For the global scale, we obtained baseline mortality data for 193 countries for 2008 from WHO mortality database (2011). It gathers mortality data reported annually by Member States from their civil registration systems. The extracted data contain yearly country-specific estimates as number of deaths, death rates and age-standardized death rates for two age-groups (15-59 and older than 60) and are stratified by cause of death: “all-cause” (ICD10); cardiovascular (ICD10: I00-I99); respiratory (ICD10: J30-J98). The European baseline data are obtained from the same set as global data.

For IdF, the mortality data for cardiovascular and respiratory mortality for 1300 communes in the Ile-de-France area, for years 2000-2010, were obtained from the CapiDC (France).

Initially, for 2030 and 2050, we kept baseline mortality data and population data constant. In sensitivity analysis we examined the projected changes in baseline mortality rates and population for 2030 in several world regions, using WHO projections (Mathers & Loncar, 2006).

## 2.5. HIA Method

HIA computes the changes in health outcomes related to changes in air-pollutant concentrations using a concentration-response function (CRF) derived from epidemiological studies. The changes in mortality are estimated using the following equation:

$$\Delta Y = Y_0 * (1 - \exp(-\beta * \Delta X)), \text{ where}$$

$\Delta Y$  is the change in incidence rate of the health effect in a given population in the future relative to present,  $Y_0$  is the baseline mortality for a given population,  $\beta$  is the CRF that describes the relationship between a change in the pollutant concentration and the corresponding change in the outcome of the health effect,  $\Delta X$  is a change in the air-pollutant concentration.

Considering the current evidence on the health impacts of OAP and health data availability, we decided to focus on the long-term impacts of  $PM_{2.5}$  and cardiovascular mortality, and of  $O_3$  on respiratory mortality. For PM, we used the CRF provided by a recent meta-analysis (Hoek et al. 2013). For ozone, we used the relative risk of respiratory disease, 1.040 [95% CI: 1.010, 1.067] per  $20 \mu g/m^3$  increase in the summer season

average of 1-hour maximum ozone converted to the RR value per  $10 \mu\text{g}/\text{m}^3$  increase in the maximum daily 8-hour ozone mean of the summer season using a 8-hour/1-hour ratio of 0.88 from Jerrett et al. (2009).

We computed  $\Delta X$  as the difference between the concentrations observed for the future under a given scenario and the 2010 period concentrations.

To produce gridded maps we calculated HIA using projected air pollution for each cell ( $1^\circ \times 1^\circ$  resolution at the global scale,  $4\text{km} \times 4\text{km}$  at the European scale, and  $0.4\text{km} \times 0.4\text{km}$  at the IdF scale). For presentation purpose, we aggregated the results into the region, country or departement levels using population-weighted air pollution calculated separately for each region of interest at all scales. The countries and regions were chosen according to the WHO classification. Gridded results are presented in the maps and can be requested from the authors.

### 3. Results

#### 3.1. Particulate Matter and Ozone Concentrations

##### 3.1.1. Global Scale

Global  $\text{PM}_{2.5}$  concentrations for 2010, 2030 and 2050 under two emissions scenarios are shown in Figure S1. For 2010, the population-weighted global average annual  $\text{PM}_{2.5}$  concentrations across the countries were estimated at  $4.0$  (min= $0.3$ , max= $22.6$ )  $\mu\text{g}/\text{m}^3$ . This value is lower than measured concentrations in several locations but owing to the spatial scale of the global analysis it is considered representative and comparable to other assessments (Rao et al. 2013). In 2030, this level would decrease by about  $0.2 \mu\text{g}/\text{m}^3$  under CLE and by  $1.9 \mu\text{g}/\text{m}^3$  under MFR. The future changes exhibit considerable geographic heterogeneity, as illustrated in supplemental Figure S2 for six world regions.

Future concentrations under CLE scenario are very similar to the present situation (it is not necessarily when the analysis is done at a finer spatial resolution, such as European and IdF; see sections 3.1.2 and 3.1.3), compared to MFR, under which large reductions are projected. Large part of the population will remain exposed to concentrations above the WHO guideline values ( $10 \mu\text{g}/\text{m}^3$  for  $\text{PM}_{2.5}$ ). The 34% of world population could be exposed to  $\text{PM}_{2.5} > 10 \mu\text{g}/\text{m}^3$  in 2010. In 2030, it could be 42% under CLE and 1% under MFR. These numbers would worsen giving the fact that many countries have their own standards that tend to be stricter than WHO's.

In Asia, the  $\text{PM}_{2.5}$  could increase by 18% under CLE scenario in 2050. Under the more aggressive emissions reductions scenario represented by the MFR, the average concentrations could decrease in Asia by 57% by 2030, especially in China (77% decrease).

Similarly large percentage reductions are projected for Europe and northern America in 2030 under MFR; the PM concentrations could decrease by 63% and 56%, respectively. In Latin America the concentrations could decrease by 32% under MFR. Similar results were estimated for Sub-Saharan (SS) Africa, where the largest improvements were projected in the southern regions (52% decrease) and lowest in the central part of SS Africa (13%) by 2030. However, under CLE, in some regions of Latin America (southern and tropical) and SS Africa (central and west) the concentrations are projected to continue to increase.

Global  $\text{O}_3$  concentrations for 2010, 2030 and 2050 under two emissions scenarios are shown in supplementary Figure S3. For 2010, the population-weighted global average summer  $\text{O}_3$  MDA1 concentrations across the countries were estimated at  $73$  (min= $21$ , max= $129$ )  $\mu\text{g}/\text{m}^3$ . In 2030, these levels would not change under CLE and would decrease by  $11 \mu\text{g}/\text{m}^3$  under MFR. The geographical heterogeneity is also high across the countries. In 2010, the largest  $\text{O}_3$  levels are projected in Europe and North America

regions (in each of these regions the level was  $102 \mu\text{g}/\text{m}^3$  on average across the countries), which could decrease by 17% and 16% in 2030 relative to 2010 under MFR, respectively. Under CLE, the decreases in Europe and North America in 2030 are 6% and 5%, respectively. In North Africa/Middle East, the 2010 concentrations and changes under both scenarios in 2030 are close to the European and North American changes. In Asia, in 2010, the average  $\text{O}_3$  level across the countries was  $79 \mu\text{g}/\text{m}^3$ . The future percent changes in  $\text{O}_3$  concentrations in Asia in 2030 relative to 2010 are similar to the European changes (17%). By contrast, under CLE, in Asia, North Africa/Middle East and Sub-Saharan Africa, the levels are projected to be larger by 7%, 6% and 2% in 2030 than it was in 2010, respectively.

### 3.1.2. European scale

Modeled  $\text{O}_3$  and  $\text{PM}_{2.5}$  concentrations in 2010 and 2030 under two emissions scenarios are shown in Figure S4 (Supplementary Material).

Our estimates for 2010  $\text{PM}_{2.5}$  matched existing model estimates, with the well-documented hotspots of anthropogenic particulate pollution in the Benelux, South Poland, the Po-Valley (Thunis et al. 2007). The 42% of European population is exposed to the levels of  $\text{PM}_{2.5}$  larger than  $10 \mu\text{g}/\text{m}^3$ . The future evolution of  $\text{PM}_{2.5}$  exhibits a strong reduction compared to 2010 levels. Under CLE, the main hotspots would remain visible. Under MFR, the PM levels would be significantly reduced..

Similar results as for  $\text{PM}_{2.5}$  were obtained for  $\text{O}_3$ . In 2010, the high  $\text{O}_3$  levels are projected over the Mediterranean and neighboring countries. Major pollution hotspots also stand out in Central Europe. Efficient reductions occur over most continental areas by 2050 but high  $\text{O}_3$  levels remain over sea surfaces (where increase in sunny days, temperatures and decrease of precipitation are more favorable to ozone build-up and the deposition sink is less efficient). This feature was found in particular over the Mediterranean, to such extent that  $\text{O}_3$  levels remains high over Spain, Italy, Greece and Balkan countries. Over Turkey, it is even an  $\text{O}_3$  increase that we find by 2030 and 2050 compared to present-day levels (Figure S4, Supplementary Material).

### 3.1.3. Ile-de-France scale

Modeled  $\text{O}_3$  and  $\text{PM}_{2.5}$  concentrations in 2010 and 2050 under two emissions scenarios are shown in Figure S5 (Supplementary Material).

In IdF, the 2010 population-weighted annual average  $\text{PM}_{2.5}$  concentration level ranged from 10 to  $21 \mu\text{g}/\text{m}^3$  across the region; the entire population is exposed to levels higher than WHO guideline values. The largest concentrations were projected in the city of Paris (over  $20 \mu\text{g}/\text{m}^3$ ). Future concentrations are projected to decrease relative to 2010: by 49% under CLE and by 69% under MFR. This is mainly due to the significant reductions in primary particle emissions. The reductions were stronger in the city of Paris than in rural areas due to the effective mitigation of road transport emissions (Markakis et al. 2014).

The 2010 population-weighted summer (June-August) MDA1  $\text{O}_3$  concentrations ranged from 87 to  $102 \mu\text{g}/\text{m}^3$  across the region. Lower  $\text{O}_3$  levels ( $87 - 91 \mu\text{g}/\text{m}^3$ ) over the city of Paris are due to titration by NO (road transport mainly). On average by 2050 the gridded concentrations are projected to decrease by 9% under CLE and by 13% under MFR, mostly in rural areas. In the city area of Paris, we expect increases in concentrations by 4% under CLE and by 2% under MFR, due to reduced NO titration. By contrast, in rural areas the  $\text{O}_3$  concentrations will decrease by 8.9% under CLE and by 12.4% under MFR.

### 3.2. Health Impact Assessment

There were 17 242 804 of CV deaths recorded in the world, 2 775 396 of which occur in Europe, and 154 741 in France. In 2030, under CLE, the total CV deaths in the world could increase by 6 600 (+0.04%) due to the changes in PM<sub>2.5</sub> concentrations. Under MFR, it would decrease by 1,5 million (-9%). In Europe, it would decrease by 109 000 (-4%) deaths under CLE, and 219 000 (-8%) under MFR. In France, the decrease would be of 7 100 (-5%) deaths under CLE, and 11 700 (-8%) under MFR (Table 1). Corresponding concentrations can be found in Table S1 in the Supplementary Material.

Table 1. Observed CV mortality in 2010 and expected number of CV deaths in 2030 and 2050 taken into account changes in PM<sub>2.5</sub> – Global, European and IdF scales

| Scale    | World Region  | Annual CV mortality (number of deaths) <sup>a</sup> |                            |                            |                            |                            |
|----------|---------------|-----------------------------------------------------|----------------------------|----------------------------|----------------------------|----------------------------|
|          |               | 2010<br>(baseline)                                  | 2030                       |                            | 2050                       |                            |
|          |               |                                                     | CLE<br>(95% CI:)           | MFR<br>(95% CI:)           | CLE<br>(95% CI:)           | MFR<br>(95% CI:)           |
| Global   | World (192)   | 17 243                                              | 17 249<br>(17 243, 17 266) | 15 755<br>(16 799, 14 840) | 17 489<br>(17 309, 17 682) | 15 762<br>(16 802, 14 850) |
|          | Europe (N=38) | 2 775                                               | 2 720<br>(2 760, 2 682)    | 2 670<br>(2 745, 2 598)    | 2 715<br>(2 758, 2 673)    | 2 667<br>(2 744, 2 592)    |
|          | France        | 155                                                 | 152<br>(154, 149)          | 149<br>(153, 146)          | 151<br>(154, 149)          | 149<br>(153, 145)          |
|          | IdF           | 17.7                                                | -                          | -                          | 17.2<br>(17.5, 16.8)       | 16.9<br>(17.4, 16.4)       |
|          | Paris         | 3.5                                                 | -                          | -                          | 3.4<br>(3.5, 3.3)          | 3.4<br>(3.5, 3.3)          |
| European | Europe (N=38) | 2 775                                               | 2 666<br>(2 744, 2 591)    | 2 557<br>(2 712, 2 412)    | 2 648<br>(2 739, 2 561)    | 2 549<br>(2 710, 2 400)    |
|          | France        | 155                                                 | 148<br>(153, 143)          | 143<br>(151, 135)          | 146<br>(152, 140)          | 142<br>(151, 134)          |
|          | IdF           | 17.7                                                | -                          | -                          | 16.2<br>(17.2, 15.2)       | 15.4<br>(17, 14)           |
|          | Paris         | 3.5                                                 | -                          | -                          | 3.2<br>(3.4, 3)            | 3<br>(3.4, 2.7)            |
| IdF      | IdF           | 17.7                                                | -                          | -                          | 15.6<br>(17.1, 14.3)       | 14.8<br>(16.8, 13.1)       |
|          | Paris         | 3.5                                                 | -                          | -                          | 3<br>(3.4, 2.7)            | 2.8<br>(3.3, 2.4)          |

All estimates are for population older than 15 years of age; <sup>a</sup> - \*1000.

#### 3.2.1. Global Scale

1.5 [95% CI: 0.4, 2.4] millions of CV deaths could be delayed each year in 2030 compared to 2010 under MFR scenario (Figure 2, Table S2). 84% of this benefit (nearly 1.3 million delayed deaths) would occur in Asia. The 0.82 millions (nearly 55%) of these belong in China and 0.24 million (16%) in India.

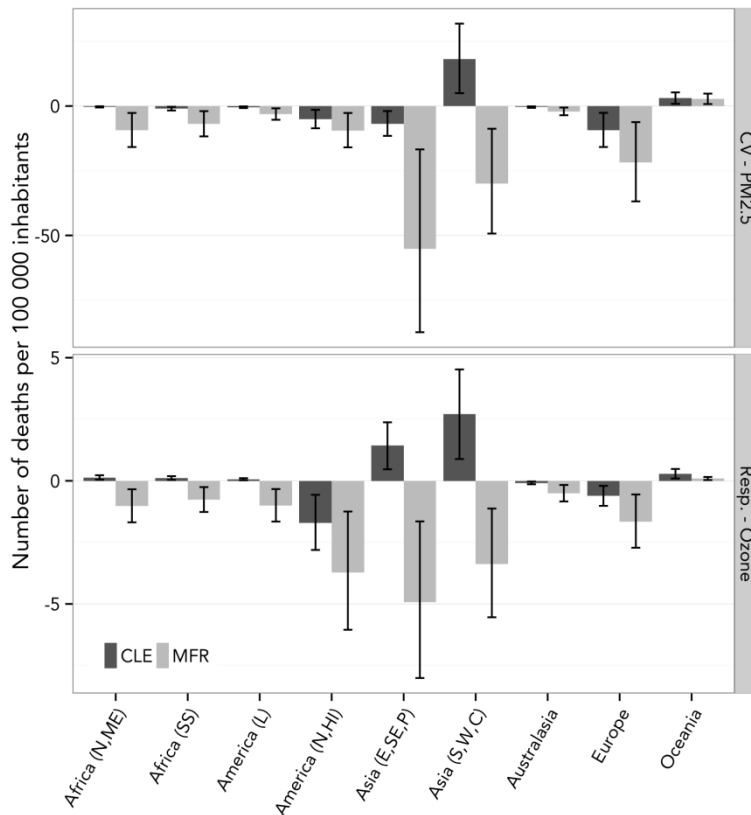


Figure 2. Predicted changes in CV and respiratory mortality per 100 000 inhabitants associated with PM<sub>2.5</sub> and O<sub>3</sub>, respectively, in 2030 under CLE and MFR scenarios including CRF limits (95% CI)

By contrast, under CLE scenario, additional CV deaths due to PM<sub>2.5</sub> would occur each year; nearly 7 000 in 2030 and 0.25 millions in 2050, compared to 2010. The largest increases in number of CV deaths are projected in South Asia (85% of total Asia increases), the largest part of which belong in India (over 0.15 millions).

In Europe, we expect decreases in CV mortality with both scenarios. Under MFR, nearly 0.14 (0.04, 0.23) millions could be delayed in 2030 compared to 2010 (9% of total world reductions) (Figure 2). The spatial repartition of impacts for 2030 is illustrated in Figure S6 (Supplementary Material).

### 3.2.2. Europe Scale

In Europe, both scenarios concluded that air pollution-related mortality should decrease in 2030 compared to 2010 (Figure S7); by nearly 8% for CV mortality, and 0.3% for respiratory mortality under MFR. The benefits under MFR scenario (219 000 CV deaths) are noticeably larger than those under CLE (109 000 CV deaths). Again, most of the changes should happen before 2030.

Under MFR, most benefits are expected in Central and Eastern Europe (especially Bulgaria, Belarus, Romania and Ukraine) for PM<sub>2.5</sub>-related deaths and in Western Europe (especially Spain, Hungary and Italy) for O<sub>3</sub>-related deaths under both, CLE and MFR scenarios (Table S3, Figure 3). These estimated changes in

CV and respiratory deaths are directly affected by the population-weighted contribution of the concentration levels in each country, that is why there is a considerable difference between Switzerland and Ukraine in PM<sub>2.5</sub>-related deaths, for example.

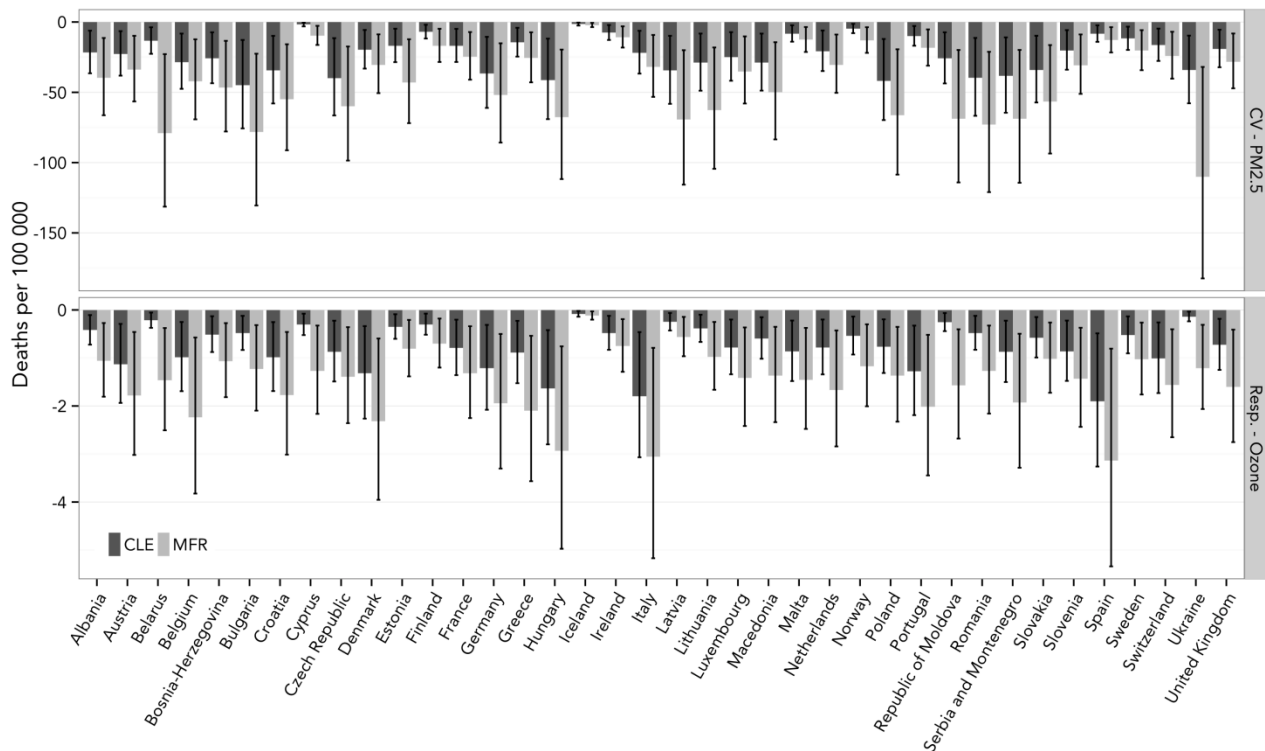


Figure 3. Predicted changes in CV and respiratory mortality per 100 000 inhabitants associated with PM<sub>2.5</sub> and Ozone in Europe in 2030 compared to 2010 under CLE and MFR scenarios (the error bars were computed with 95% confidence interval of the RR). Results by country

### 3.2.3. Ile-de-France Scale

The IdF pattern is consistent with the global and regional scales results: more reductions in deaths are expected under MFR, and the benefits associated with PM<sub>2.5</sub> changes largely exceed the benefits associated with ozone changes. More than 2 830 annual CV deaths could be delayed in IdF in 2050 compared to 2010, with a distribution of benefits across towns (Table 2, Figure 4). The larger reductions in CV mortality (14% under CLE and 20% under MFR compared to 2010) are expected in the city area of Paris, where PM<sub>2.5</sub> could decrease by more than 80% under the MFR scenario.

Table 2. Present population and annual CV mortality baseline, and future changes in cardiovascular and respiratory deaths associated with population-weighted annual PM<sub>2.5</sub> and ozone in 2050 compared to 2010 - Ile-de-France Scale

| IdF area | Number of communes | Pop., 2010 | Annual CV mortality baseline, 2010 | Future changes in CV deaths due to PM <sub>2.5</sub> |                  | Annual Resp. mortality baseline | Future changes in Resp. deaths due to O <sub>3</sub> |                  |
|----------|--------------------|------------|------------------------------------|------------------------------------------------------|------------------|---------------------------------|------------------------------------------------------|------------------|
|          |                    |            |                                    | 2050 - baseline                                      |                  |                                 | 2050 - baseline                                      |                  |
|          |                    |            |                                    | CLE<br>(95% CI:)                                     | MFR<br>(95% CI:) |                                 | CLE<br>(95% CI:)                                     | MFR<br>(95% CI:) |

|               |       |           |        |                          |                          |       |                  |                   |
|---------------|-------|-----------|--------|--------------------------|--------------------------|-------|------------------|-------------------|
| Paris         | 20    | 1 913 400 | 3 507  | -496<br>(-147, -805)     | -699<br>(-212, -1 108)   | 876   | +5<br>(+1, +9)   | +2<br>(+1, +4)    |
| Paris Suburbs | 411   | 6 482 500 | 11 802 | -1 369<br>(-402, -2 242) | -1 883<br>(-563, -3 031) | 2 812 | -26<br>(-7, -45) | -43<br>(-11, -74) |
| Rural areas   | 869   | 1 040 300 | 2 353  | -183<br>(-53, -305)      | -253<br>(-74, -416)      | 576   | -11<br>(-3, -19) | -15<br>(-4, -26)  |
| IdF total     | 1 300 | 9 436 100 | 17 662 | -2 049<br>(-602, -3 351) | -2 835<br>(-849, -4 555) | 4 264 | -33<br>(-8, -56) | -56<br>(-14, -96) |

All estimates are for population older than 15 years of age; Pop. - population counts.

For O<sub>3</sub>, we expect that the respiratory mortality will increase in the city of Paris by 0.6% under CLE and by 0.3% under MFR (relative to baseline) and decrease in rural areas (by 2.0% under CLE and by 2.7% under MFR) (Figure 4).

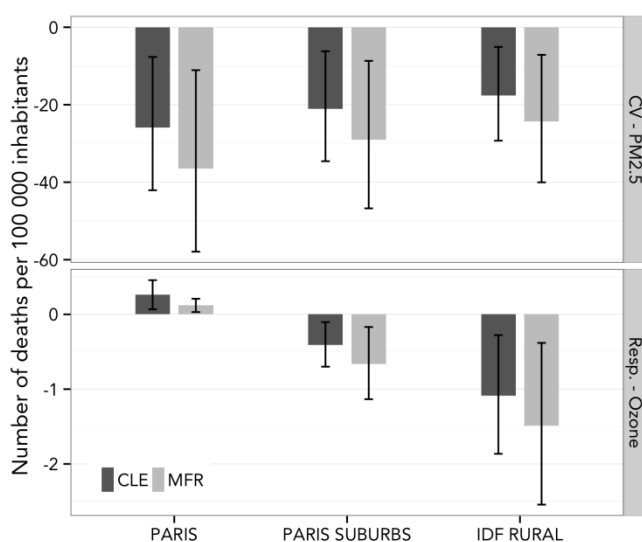


Figure 4. Predicted changes in CV and respiratory mortality per 100 000 inhabitants associated with PM<sub>2.5</sub> and Ozone in IdF in 2050 compared to 2010 under CLE and MFR scenarios (the error bars were computed with the 95% confidence interval of the RR)

### 3.3. Influence of the geographical scale: Global vs. Regional and Regional vs. Local

In this study the global and regional models use identical air pollutant emissions. The INCA global model has been recently improved to include chemistry of secondary inorganic aerosols, becoming closer to regional air quality models. The only major discrepancy in model formulation is the lack of secondary organic aerosol in the global model. In addition, model resolution may have a significant impact on HIA, as a finer scale allows a better assessment of population exposure.

For the same geographical domain area, the 2010 PM<sub>2.5</sub> concentrations modeled at the European scale were about twice as large as the global scale PM<sub>2.5</sub>. The slope of the relationship between European and global scale was 1.69 ( $r^2=0.76$ ) and remained almost unchanged in the future concentrations (Figure 5, Figure S8). This discrepancy in concentrations, which is caused mostly by the different model use, increased

the discrepancy in CV mortality to 2.32 ( $r^2=0.98$ ). This factor increased when we estimated present burden of PM<sub>2.5</sub> on CV mortality using concentrations over 3  $\mu\text{g}/\text{m}^3$ , to 4.64 ( $r^2=0.86$ ), that is about 246 000 CV deaths at the European scale vs. 45 000 CV deaths at the global scale (Figure 5). The 3  $\mu\text{g}/\text{m}^3$  threshold is the minimum concentration (that we could find) used in the cohort studies, from which Hoek et al. (2013) pooled the relative risks for cardiovascular mortality. The ratio between the 2010 PM<sub>2.5</sub> reduced to the threshold and total PM<sub>2.5</sub> concentrations was on average 1.5 for the European scale estimates and 4.2 for the global scale.

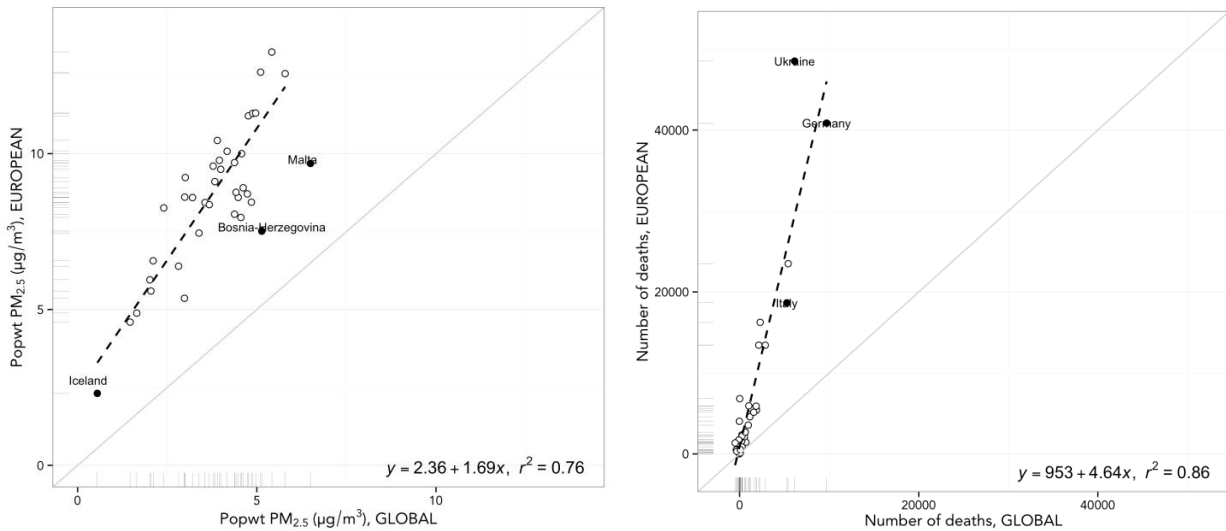


Figure 5. Predicted population-weighted PM<sub>2.5</sub> concentrations (left) and CV mortality associated with PM<sub>2.5</sub> (right) in 2010: Comparison of the estimates obtained at the global and European scales for 38 countries in Europe. The red line indicates the linear fit between the global and European scales estimates. The solid points indicate the outliers according to Cook's distance

The discrepancy between global and European scales in PM<sub>2.5</sub> changes in 2030 or 2050 relative to 2010 under both scenarios was about 1.5; here, the discrepancy could also be affected by the difference in climate evolution used in the two models. Consequently, reductions in CV deaths projected with the global model are less than half of those obtained with the European scale (Table 3, Figure S9).

Table 3. Predicted changes in population-weighted PM<sub>2.5</sub> concentrations and associated CV deaths in 2030 and 2050 compared to 2010 in Europe (N=38 countries) at the global and European scales

| Health outcome                        | Year | Scenario | European countries at the Global Scale<br>(N <sub>countries</sub> = 38) |                                                |                      | European countries at the European Scale (N <sub>countries</sub> = 38) |                                                |                      |
|---------------------------------------|------|----------|-------------------------------------------------------------------------|------------------------------------------------|----------------------|------------------------------------------------------------------------|------------------------------------------------|----------------------|
|                                       |      |          | Conc. range, $\mu\text{g}/\text{m}^3$                                   | Avr. change in conc., $\mu\text{g}/\text{m}^3$ | Changes in CV deaths | Conc. range, $\mu\text{g}/\text{m}^3$                                  | Avr. change in conc., $\mu\text{g}/\text{m}^3$ | Changes in CV deaths |
|                                       |      |          |                                                                         |                                                |                      |                                                                        |                                                |                      |
| CV mortality due to PM <sub>2.5</sub> | 2030 | CLE      | 0.5 - 5.7                                                               | -1.3                                           | -55 099              | 2.2 - 9.0                                                              | -2.4                                           | -109 398             |
|                                       |      | MFR      | 0.5 - 4.8                                                               | -2.6                                           | -105 356             | 2.0 - 7.5                                                              | -5                                             | -218 843             |
|                                       | 2050 | CLE      | 0.5 - 5.6                                                               | -1.4                                           | -60 538              | 1.9 - 8.4                                                              | -2.5                                           | -127 620             |

|  |  |     |           |      |          |           |      |          |
|--|--|-----|-----------|------|----------|-----------|------|----------|
|  |  | MFR | 0.5 - 4.6 | -2.9 | -108 613 | 1.8 - 7.3 | -5.1 | -226 523 |
|--|--|-----|-----------|------|----------|-----------|------|----------|

All estimates are for population older than 15 years of age. The changes in CV mortality are relative to the present baseline mortality, which was 2 775 596 deaths annually.

For ozone, the global and European scales future estimates generally agree for almost all countries in Europe (Figure S10, Supplementary Material).

Assuming that this relationship can be extrapolated to other world regions, we estimated the underestimation of PM<sub>2.5</sub>-related CV deaths derived from the global scale modeling. After applying the correction model, we computed that 329 300 additional PM<sub>2.5</sub>-related CV deaths could occur in 2030 compared to 2010, which is 50 time larger than the initial global scale estimates. Benefits under MFR would increase by more than 1 million CV-related deaths (Table 4).

Table 4. Present population, annual CV mortality baseline and changes in CV mortality associated with population-weighted PM<sub>2.5</sub> in 2030 compared to 2010 - Comparison of the global scale estimates (base) and corrected global scale estimates (fitted)

| World Region        | Number of countries | Pop., <sup>a</sup> 2010 | Annual CV mortality baseline, <sup>a</sup> 2010 | Changes in CV deaths <sup>a</sup> , 2030 CLE - baseline |                            | Changes in CV deaths <sup>a</sup> , 2030 MFR - baseline |                                |
|---------------------|---------------------|-------------------------|-------------------------------------------------|---------------------------------------------------------|----------------------------|---------------------------------------------------------|--------------------------------|
|                     |                     |                         |                                                 | base (95% CI:)                                          | fitted (95% CI:)           | base (95% CI:)                                          | fitted (95% CI:)               |
| Africa (N,ME)       | 19                  | 315 899                 | 926                                             | -1.0<br>(-0.3, -1.8)                                    | -5.5<br>(-1.6, -9.4)       | -29.6<br>(-8.4, -50.1)                                  | -64.0<br>(-18.4, -106.6)       |
| Africa (SS)         | 46                  | 469 674                 | 1 295                                           | -4.7<br>(-1.3, -8.0)                                    | -6.7<br>(-1.9, -11.3)      | -32.5<br>(-9.2, -55.1)                                  | -65.1<br>(-18.7, -109.0)       |
| America (L)         | 33                  | 407 461                 | 989                                             | -1.9<br>(-0.5, -3.2)                                    | -9.9<br>(-2.8, -16.9)      | -12.7<br>(-3.6, -21.6)                                  | -35.2<br>(-10.0, -59.3)        |
| America (N,HI)      | 2                   | 275 647                 | 948                                             | -13.9<br>(-3.9, -23.7)                                  | -30.5<br>(-8.7, -51.6)     | -26.0<br>(-7.4, -44.0)                                  | -57.6<br>(-16.5, -96.3)        |
| Asia (E,SE,P)       | 19                  | 1 668 655               | 5 568                                           | -114.1<br>(-32.5, -192.3)                               | +27.6<br>(+7.4, +49.3)     | -920.3<br>(-279.6, -1 458.9)                            | -1 459.0<br>(-467.1, -2 207.3) |
| Asia (S,W,C)        | 14                  | 1 101 916               | 3 393                                           | +200.4<br>(+54.9, +351.1)                               | +476.5<br>(+126.3, +862.9) | -330.6<br>(-96.7, -543.1)                               | -565.4<br>(-170.4, -903.2)     |
| Australasia         | 3                   | 20 400                  | 60                                              | -0.1<br>(-0.0, -0.1)                                    | -0.7<br>(-0.2, -1.2)       | -0.4<br>(-0.1, -0.7)                                    | -1.7<br>(-0.5, -2.8)           |
| Europe              | 43                  | 623 945                 | 4 050                                           | -58.2<br>(-16.5, -98.8)                                 | -121.6<br>(-34.6, -205.1)  | -136.2<br>(-38.7, -230.0)                               | -286.9<br>(-82.7, -477.6)      |
| Europe              | 38                  | 498 142                 | 2 775                                           | -55.1<br>(-15.6, -93.5)                                 | -108.9<br>(-31.0, -183.5)  | -105.4<br>(-30.0, -177.6)                               | -215.7<br>(-62.3, -358.3)      |
| Europe <sup>b</sup> | 38                  | 498 142                 | 2 775                                           | -109.4<br>(-31.2, -184.1)                               | -                          | -218.8<br>(-63.3, -363.1)                               | -                              |
| Oceania             | 13                  | 5 349                   | 14                                              | +0.2<br>(+0.0, +0.3)                                    | +0.2<br>(+0.1, +0.4)       | +0.1<br>(+0.0, +0.3)                                    | +0.1<br>(+0.0, +0.2)           |

|                    |     |           |        |                       |                           |                                |                                |
|--------------------|-----|-----------|--------|-----------------------|---------------------------|--------------------------------|--------------------------------|
| World <sup>c</sup> | 192 | 4 888 946 | 17 243 | +6.6<br>(-0.1, +23.4) | +329.3<br>(+84.1, +617.1) | -1 488.3<br>(-443.7, -2 403.2) | -2 534.7<br>(-784.3, -3 962.0) |
|--------------------|-----|-----------|--------|-----------------------|---------------------------|--------------------------------|--------------------------------|

All estimates are for population older than 15 years of age. Pop. - population counts; <sup>a</sup> - \*1000; <sup>b</sup> - European scale estimates; <sup>c</sup> - the total do not include European estimates for 38 countries; N – north; ME – Middle East; SS – Sub-Saharan; L – Latin; HI – High Income; E – East; SE – South-East; P – Pacific; S – South; W – West; C – Central; the highlighted rows are not included in the world's total estimate.

For IdF domain, the IdF scale PM<sub>2.5</sub> concentrations were larger than those derived from the European scale. A comparison between these two scales can be found in Supplemental Material (Figure S11, Table S4). In that case the larger differences were obtained for ozone. Both scales mostly agree on their modeled concentrations in the IdF totals, because the results are dominated by the rural areas, which cover the majority of the IdF domain. When we separate the highly populated city of Paris from the rural areas, the results for Paris show larger discrepancy; increases in O<sub>3</sub> concentrations at the IdF scale and decreases at the European scale. Consequently, impacts of ozone changes on respiratory mortality over Paris were 15 time larger at the IdF scale rather than the European scale, under MFR. Indeed, the two scales used different scaling methodologies for emissions and due to the resolution constraints at the European scale the ozone chemistry inside the city cannot be captured with the same precision as it is at the IdF scale (see Section 3.1.3).

## 4. Discussion and Conclusions

### 4.1. Main results

We used the same methodology on three different geographical scales to estimate the long-term future effects of PM<sub>2.5</sub> on cardiovascular mortality and O<sub>3</sub> on respiratory mortality. Keeping consistent methodology across the scales is a hard task, especially because of the lack of consistent datasets and emissions scenarios among scales. The interdisciplinary approach complements the understanding of processes from different points of view and provides shared access to various methods that can help to achieve the common goal.

On every scale, large public health benefits are expected under MFR, while the burden of AP on mortality could worsen in many places under CLE. Yet, some countries, especially in Africa, would not benefit from an air quality improvement even under MFR.

In Europe, under MFR the CV deaths can be reduced by nearly 219 000 deaths and respiratory deaths by 109,000 deaths in 2030 relative to 2010. In 2050, in some regions, climate change would have a negative impact and counterbalance the benefits obtained in 2030 because of air pollution policies.

The large numbers obtained under MFR only quantify a small part of the overall health benefits associated with air pollution improvement, since we could not quantify the benefits on morbidity or on the prevalence of some chronic diseases for instance. MFR scenario is considered unrealistic because of its costs (TSAP 2013). However, public health benefits should also be taken into account and are clearly in favor of implementing the most ambitious scenario. It has to be noted that MFR only relies on technical solutions to reduce emissions. Other solutions can be adopted from the studies on behavior changes, such as promoting active transport in cities, which has been showed to generate large co-benefits through a general improvement in physical and mental health. Such strategies would reduce GHG emissions and therefore climate change. Although the long-term benefits of reducing the intensity of climate change cannot be quantified, they are believed to be huge; climate change is one of the greatest threats, which can affect health

through multiple pathways (extreme events, modifications of the environment, infectious diseases, nutrition, risk of conflicts, etc.) (The Lancet Series 2009).

#### 4.2. Strength and Limitations

This study used air pollution emissions scenarios (ECLIPSE), which are more suitable for air pollution studies than RCPs. Although RCPs include air pollutants, they were designed exclusively for the purpose of radiative forcing investigations as part of the Atmospheric Chemistry and Climate Model Intercomparison Project (ACCMIP) project (Shindell et al. 2012, Young et al. 2012). Their implementation for air quality projections constitutes a deviation of their primary purpose (Butler et al. 2012, Colette et al. 2012b, Fiore et al. 2012). Using scenarios designed for air quality provide a better quantitative view into the efficiency of air quality policies.

Regarding the robustness of the results, the global model (LMDz-INCA) has been extensively compared to measurements for both, ozone and particles (Hauglustaine et al. 2014, Folberth et al. 2006, Wang et al. 2014), and has been used in several model inter-comparison exercises (AEROCOM, ACCMIP, HTAP). All these exercises indicate that LMDz-INCA behaves properly and provides results in agreement with other state-of-the-art global models (the list of references can be provided). Similarly to the global model, the regional (CHIMERE) model has been extensively compared to other tools and observation (e.g. the Eurodelta, AQMEII, CityZen projects) (Bessagnet et al. 2014, Galmarini et al. 2012, Colette et al. 2011, Thunis et al. 2007). For the specific purpose of future projections, the present setup would benefit from using an ensemble approach (using several models for each scale). The goal of our project was to put together a single suite of models, but in the future uncertainties and robustness can only be addressed by using ensemble of models as a regular practice. On the local scale, the large uncertainties are associated with the implemented model resolution including that of the input emissions inventory, which is particularly important over urban areas.

We performed several sensitivity analyses on each scale and compared the influence of scale on results keeping other parameters constant. We found that HIA results for Europe derived from the global scale were nearly 50% lower than those based at the European scale model. This indicates that our global scale results may be largely under-estimated.

Indeed, regional scale models can account for a shorter-term variability of pollution in space and incorporate more complex chemical schemes than global models. They use inventories and trends designed for such regional scales therefore they are more relevant for regional climate and air-quality policymaking. Also the results of regional modeling can often be compared across nearby countries. However, using relatively coarse horizontal resolutions (e.g., 50 km grid) in regional scale applications may be unsuitable when describing air pollution in urban agglomerations with high population density. It must be noted that while high model resolution can resolve transport and chemistry more efficiently, in order to describe city-scale trends, their emissions scenarios should also be developed by implementing a bottom-up approach utilizing activity information at such urban scale. In Markakis et al. (2014), where emission fluxes are taken from a local inventory to model air pollution in the area integrating only local scale data it was shown that the emission input can have a profound impact in modeled air quality compared to regional scale assessments implementing coarse inventories and resolutions.

Likewise, working at the IdF scale, the results were more than 20% larger than those estimated at the European scale for the same area of interest. Here the discrepancy can be mainly due to the resolution change. The differences were much smaller compared to the discrepancy between the European and global scales. This is partly because the same model was used at the European and IdF scales (CHIMERE). But it could also be because the gap in model resolution becomes less sensitive between the regional and local scales when focusing on the domain-wide totals, which represent the rural areas. We have shown that the

change in resolution can have a profound effect when focusing on areas with large emission gradients such as cities (Markakis et al. 2014).

For ozone, the discrepancy between the scales is larger than for PM. The European scale reductions are larger of those at the IdF scale by more than 60%. Also, by increasing the resolution over the city area of Paris the future ozone concentrations and the related deaths at the IdF scale increased in contrast with the European scale where the concentration decreased. Consistently, Thompson et al. (2012) performed uncertainty analysis using different resolutions (from 36km to 2km) and they found lower reductions in modeled ozone concentrations at a finer resolution (4-2km) than it was at more coarse one (36km) over several large U.S. cities.

Using future population and baseline mortality projections instead of keeping them constant reduced the benefits obtained with MFR scenario and enhanced the negative impact under CLE scenario for most of the world regions (Table S5). Using constant population and baseline mortality, in 2030 in Europe, under MFR scenario, the number of CV deaths could be reduced by 105 400 (95%CI: -30 000, -177 600) deaths at the global scale and by 218 800 (95%CI: -63 300, -363 100) deaths at the European scale. Taken into account changes in population and baseline mortality in the future, the CV deaths could be reduced by 185 900 (95%CI: -53 800, -308 500) deaths under the MFR scenario at the European scale, 54% of which belong to females and 46% to males (Figure 6).

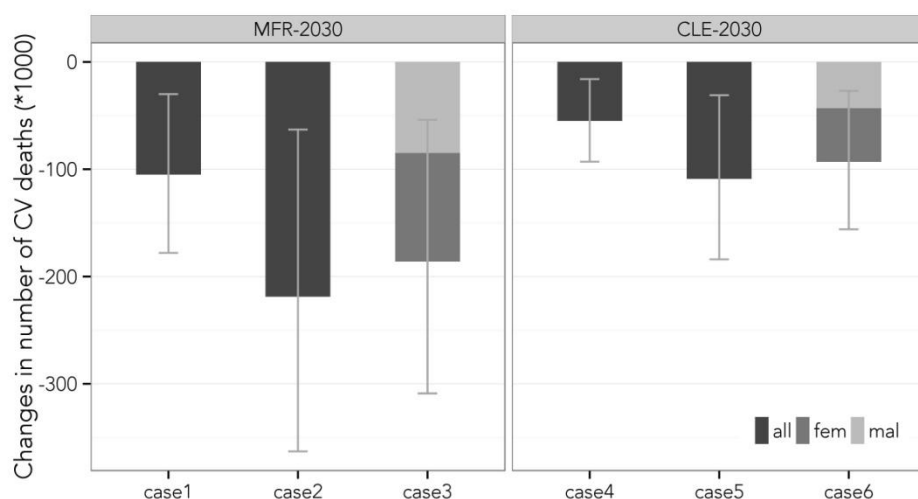


Figure 6. CV mortality associated with PM<sub>2.5</sub> in Europe in 2030: Uncertainties due to the model choice (LMDz-INCA, CHIMERE), scenario (MFR, CLE), and the use of future baseline mortality and population including 95% CI. Cases 1 and 4: LMDz-INCA model, constant mortality baseline and population; Cases 2 and 5: CHIMERE model, constant mortality baseline and population; Cases 3 and 6: CHIMERE model, changed mortality baseline and population.

Our choice of CRF might not be representative for all developing countries, but it is sufficient in our case; there were no robust long-term (cohort based) studies of PM<sub>2.5</sub>-CV and ozone-respiratory RRs from those regions. Because the range of exposure to PM<sub>2.5</sub> from diverse sources can vary widely in time and space, the relation between PM<sub>2.5</sub> exposure and excess mortality RR is not necessary to be described as linear (Pope et al. 2009, 2011; Burnett et al. 2014). However, because we did not obtain PM<sub>2.5</sub> values higher than 55 µg/m<sup>3</sup> and the pattern of the non-linear relationship between CV mortality and PM<sub>2.5</sub> is not known (so it is between respiratory mortality and O<sub>3</sub>), we used linear function instead of, for example, non-linear IER function (Burnett et al. 2014). Also, considering that only 5% of the world population is exposed to the values greater than 30 µg/m<sup>3</sup> the pattern of the relationship between CV mortality and PM<sub>2.5</sub> could be described as almost linear.

All studies to-date assume that the CRFs observed today will still be valid in the future. The assumption of constant baseline rates and constant CRFs does not take into account changes in the health-care system or in the vulnerability of the population, including aging, nor potential changes in the composition of future air-pollution mixtures. In the near future more CRFs should be available for more pollutants, including PM components, and new air-quality models will provide data for multiple PM components. Specifically, different components of PM may result in differing health impacts. With a careful methodology accounting for the possible interactions between the components, their effects on health can be better identified. And using multi-pollutant modeling approaches can provide useful uncertainty estimates.

#### 4.4. Perspectives

HIAs exploring the joint effect of climate and air-pollution policies are of particular interest to policy-makers as they can illustrate the direct consequences of costly policies. An effort to communicate effectively with policy makers on the impacts of air quality emissions' policies and of mitigation policies on air pollutants and on greenhouse gases, and their joint impacts on health, is urgently needed.

The projection of future health impacts of air pollution may play an important role in policy discussions on air pollution and climate mitigation by illustrating the health effects of alternative air pollution scenarios and of mitigation policies, and the possible negative influence of climate change on air pollution. Consistent set of methods for carrying out such projections has not yet emerged and a challenge for future HIAs is to better understand and describe the uncertainties and public-health implications of the different choices made for the exposure modeling and scenarios.

HIAs provide results that may help decision-makers choose between different policy alternatives at different scales. Global scale studies include by design chemical mechanism that represent large-scale chemical interactions relevant for international climate policy. They provide comprehensive information across multiple continents and countries and are likely to be relevant in the context of global negotiations over greenhouse gas control policies. However, their capacity to represent local to regional air-quality policy is limited because of data quality control, data comparability across countries and scalability, and their HIAs are likely to be underestimated.

One important input for a policy decision is an explicit knowledge about the level of uncertainty of the information they get. The quantitative techniques used in the HIA analysis have the potential to provide the confidence levels ascribed to the results so that they can be taken into account by policy-makers in their own quantitative risk estimates. In this study we were able to propose a first view into the uncertainties arising from using multiple climate/air quality models, climate/emissions change scenarios, and multiple geographical scales.

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